Lecture #12 | Machine learning (cont.)

SE377 Introduction to Big Data Analysis and Visualization (2017)

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Course schedule

- Project proposal: 11/9
- Machine Learning: ~11/16
- Web scraping: ~11/30
- Other topics (tentative): ~12/7
  - Regular expression
  - Text analysis
  - Introduction of map reduce
- Final presentations: 12/11~12/15
- Note: we may have guests during the class on 11/23

Recap: types of machine learning

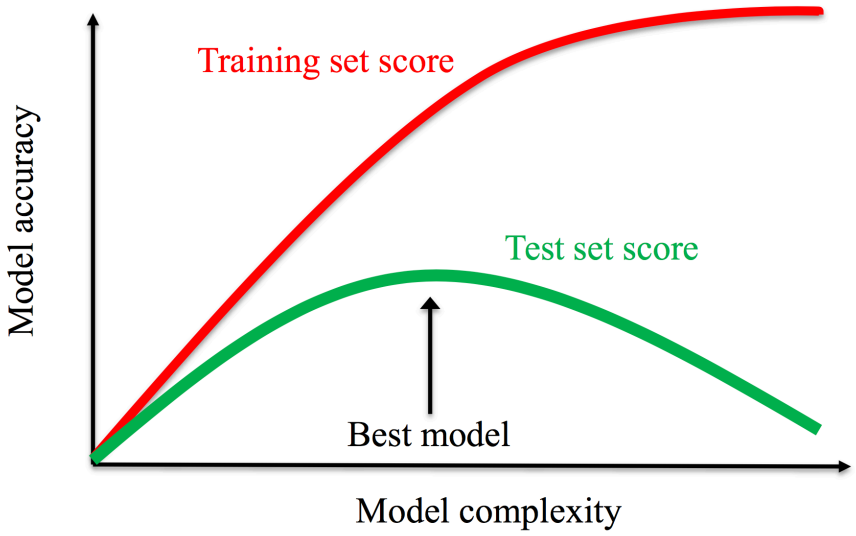
- Supervised Learning
  - Training data is labeled
  - Goal is correctly label new data
  - Examples: **classification (for categorical data), regression (for continuous data)**
  - c.f., semi-supervised learning
- Reinforcement Learning
  - Training data is unlabeled
  - System receives feedback for its actions
  - Goal is to perform better actions
- Unsupervised Learning
  - Training data is unlabeled
  - Goal is to categorize the observations
  - Examples: clustering, outlier detection

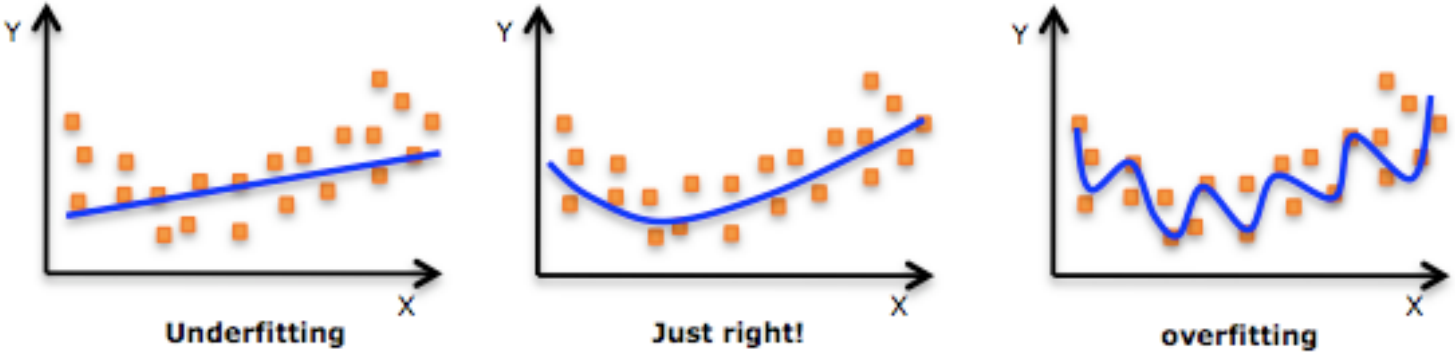
Supervised Learning Algorithms

- K-Nearest Neighbor
- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines
- Neural Networks

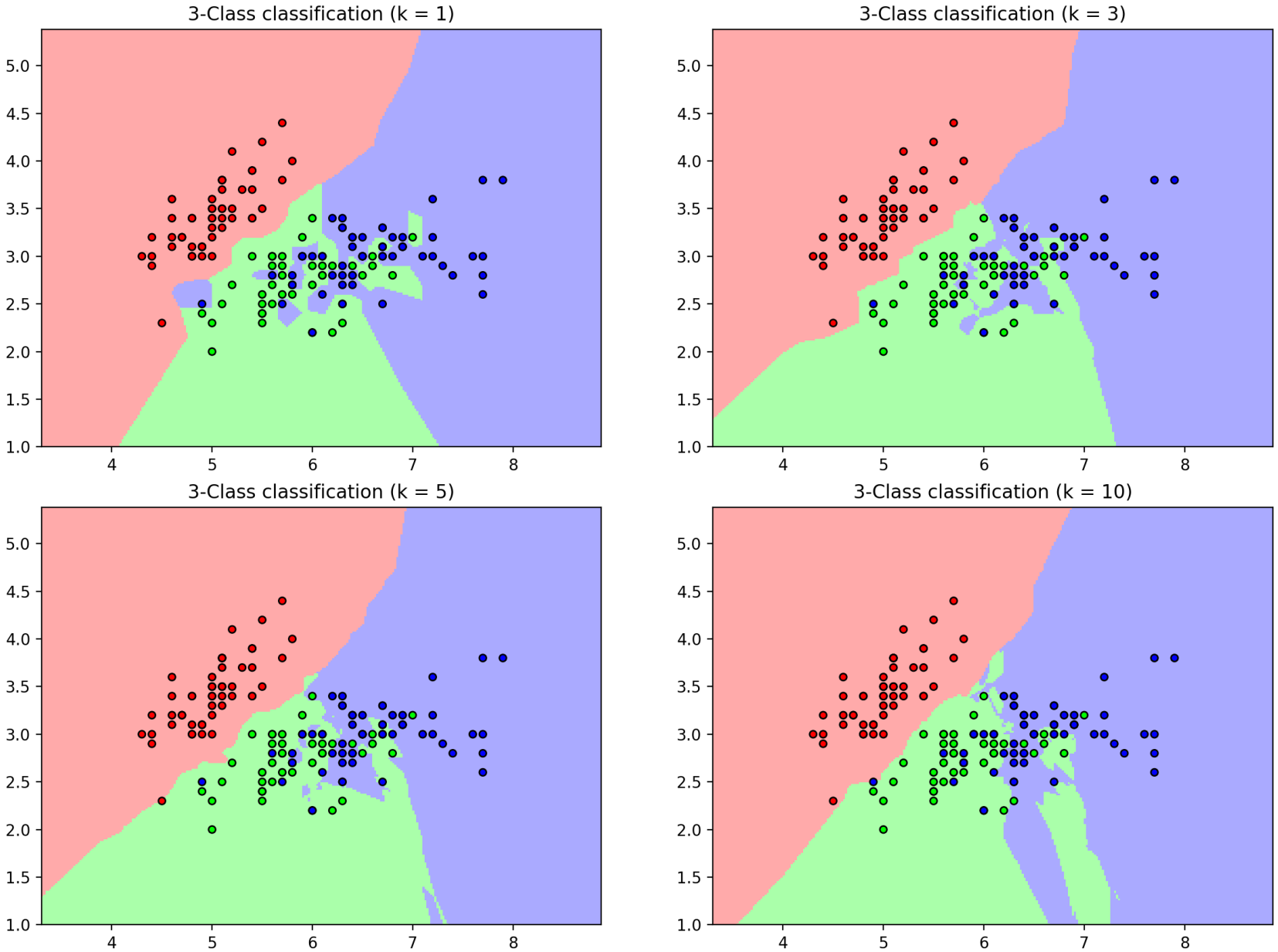
Recap:Training and test data

- Training Data: data used to learn a model
- Test Data: data used to assess the accuracy of model
- Overfitting: model performs well on training data but poorly on test data

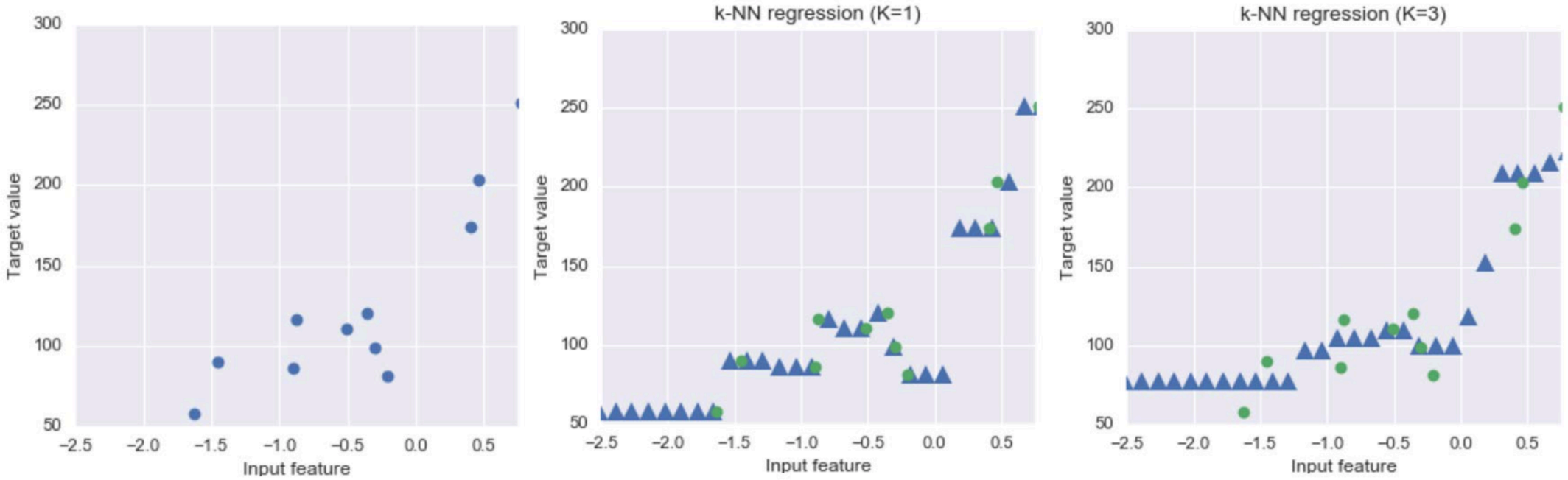




Visual representation of k-NN

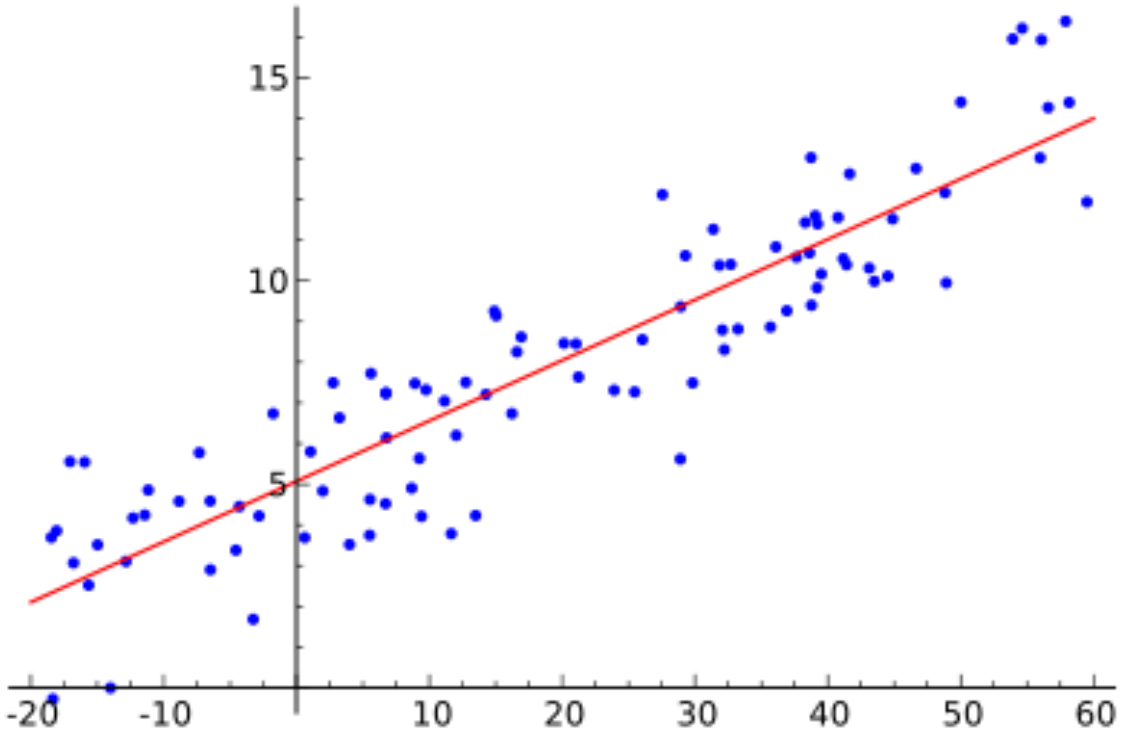


k-Nearest Neighbors Regressions



Linear regressions

- Find the best line (linear function y=f(x)) to explain the given data



General linear model

- Linear regression is an example of a linear model
- Input: $\boldsymbol{x} = (x_0, x_1, \cdots, x_n)$
- Predicted output: $\hat{y} = \widehat{w_0} x_0 + \widehat{w_1} x_1 + \cdots + \widehat{w_n} x_n$
- Parameters to estimate
  - $\hat{w} = (\widehat{w_0}, \widehat{w_1}, \cdots, \widehat{w_n})$: feature weights/ model coefficients
  - $\hat{b}$: constant bias term/intercept

Ridge regression

- Ridge regression learns w, b using the same least-squares criterion but adds a penalty for large variations in w parameters

$$RSS_{RIDGE}(\boldsymbol{w}, b) = \sum_{i=1}^{N} \left(\boldsymbol{y}_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b)\right)^2 + \alpha \sum_{j=1}^{p} w_j^2$$

- Once the parameters are learned, the ridge regression prediction formula is the same as ordinary least-squares.
- The addition of a parameter penalty is called ***regularization***. Regularization prevents overfitting by restricting the model, typically to reduce its complexity.
- Ridge regression uses L2 regularization: minimize sum of squares of w entries
- The influence of the regularization term is controlled by the parameter.
- Higher alpha means more regularization and simpler models

Feature Normalization

- Important for some machine learning methods that all features are on the same scale (e.g. faster convergence in learning, more uniform or 'fair' influence for all weights)
  - e.g. regularized regression, k-NN, support vector machines, neural networks,
- Can also depend on the data. For now, we do MinMax scaling of the features:
  - For each feature x_i: compute the min value x_i^{MIN} and the max value x_i^{MAX} achieved across all instances in the training set
  - For each feature: transform a given feature x_i value to a scaled version $x_i{}'$ using

$$x_i' = \frac{x_i - x_i^{MIN}}{x_i^{MAX} - x_i^{MIN}}$$

Lasso regression

- L1 penalty: Minimize the sum of the ***absolute values*** of the coefficients

$$RSS_{LASSO}(\boldsymbol{w}, b) = \sum_{i=1}^{N} \left(\boldsymbol{y}_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b)\right)^2 + \alpha \sum_{j=1}^{p} |w_j|$$

- This has the effect of setting parameter weights in w to zero for the least influential variables. This is called a sparse solution: a kind of feature selection
- The parameter controls amount of L1 regularization (default = 1.0).
- The prediction formula is the same as ordinary least-squares.
- When to use ridge vs lasso regression
  - Many small/medium sized effects: use ridge
  - Only a few variables with medium/large effect: use lasso

Polynomial Features with Linear Regression

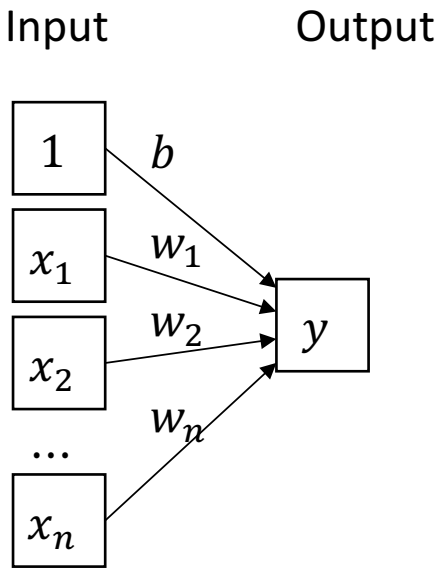
$$\boldsymbol{x} = (x_0, x_1) \rightarrow \boldsymbol{x}' = (x_0, x_1, x_0^2, x_0 x_1, x_1^2)$$

$$\hat{y} = \widehat{w_0} x_0 + \widehat{w_1} x_1 + \widehat{w_{00}} x_0^2 + \widehat{w_{01}} x_0 x_1 + \widehat{w_{11}} x_1^2 + b$$

- Generate new features consisting of all polynomial combinations of the original two features (x_0, x_1)
- The degree of the polynomial specifies how many variables participate at a time in each new feature (above example: degree 2)
- This is still a weighted linear combination of features, so it's still a linear model, and can use same least-squares estimation method for w and b

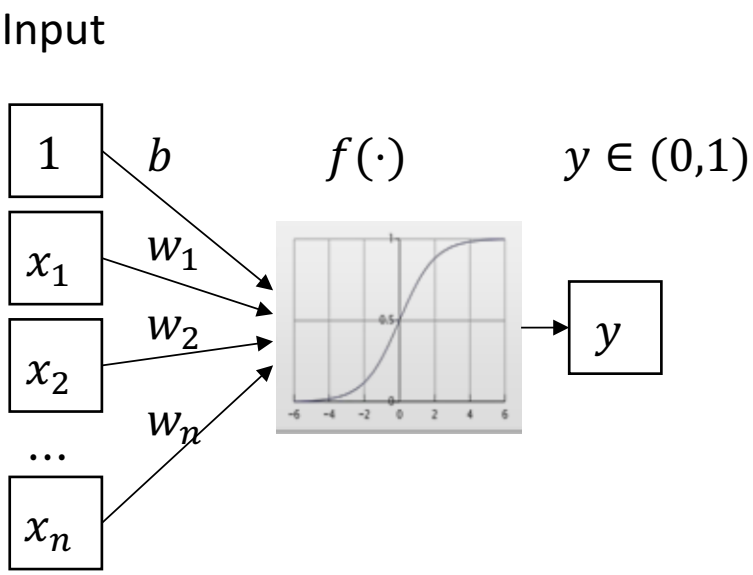
Logistic regression: comparison w/linear regression

- Linear Model



$$\hat{y} = \hat{b} + \widehat{w_1} \cdot x_1 + \cdots + \widehat{w_n} \cdot x_n$$

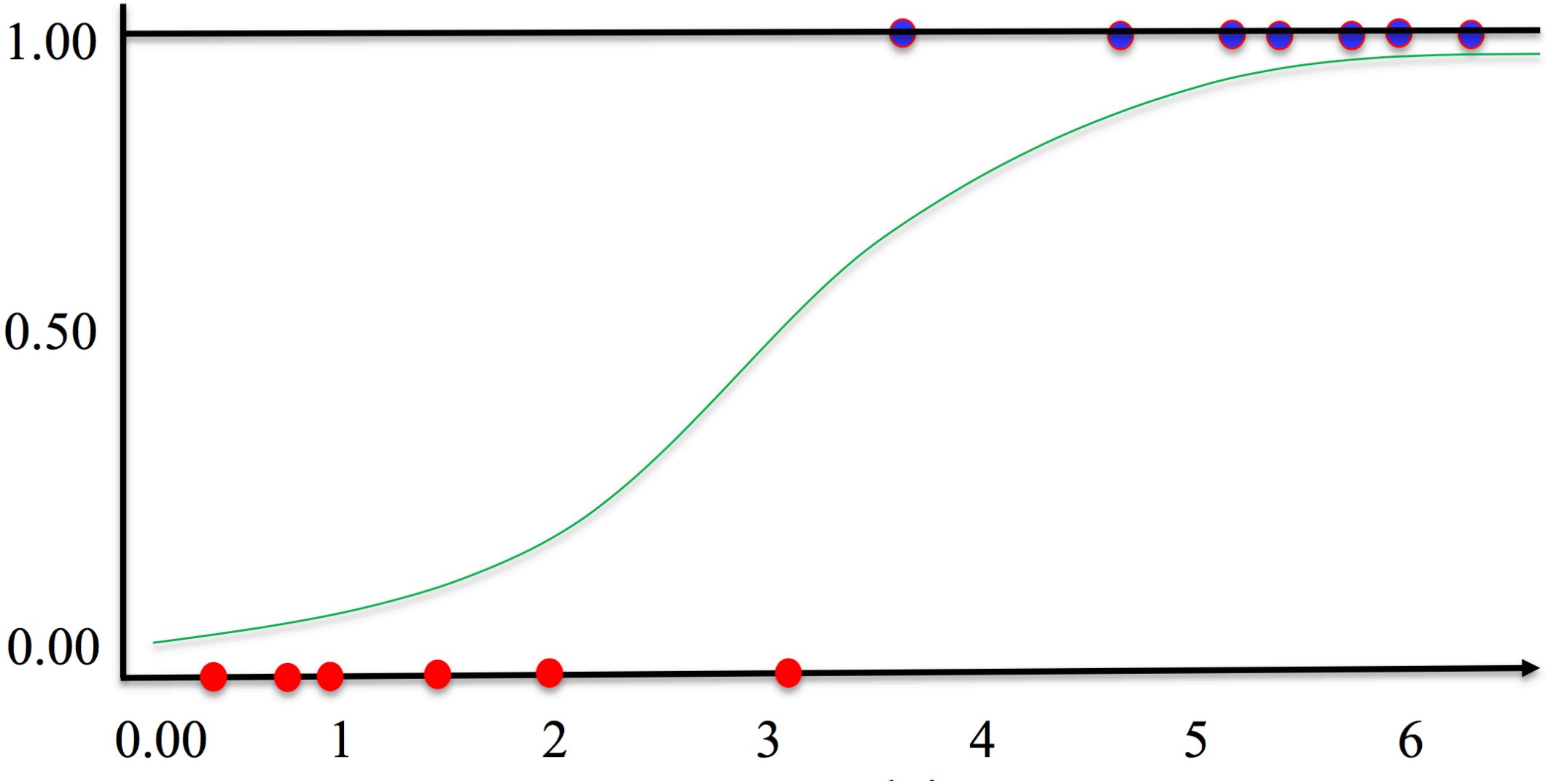
- Logistic regression: the logistic function transforms real-valued input to an output number y between 0 and 1, interpreted as the ***probability*** the input object x belongs to the positive class



$$\hat{y} = \text{logistic}\left(\hat{b} + \widehat{w_1} \cdot x_1 + \cdots + \widehat{w_n} \cdot x_n\right)$$

$$= \frac{1}{1 + \exp\left[-\left(\hat{b} + \widehat{w_1} \cdot x_1 + \cdots + \widehat{w_n} \cdot x_n\right)\right]}$$

Example: logistic regression



Maximum margin linear classifier: Linear SVM*

$$x \longrightarrow \boxed{f} \longrightarrow y$$

$$f(x, w, b) = sign(w \circ x + b)$$

- Classifier margin: the maximum width the decision boundary area can be increased before hitting a data point
- Maximum margin classifier: the linear classifier with maximum margin is a linear support vector machine (LSVM)

* Support Vector Machine

Regularization for SVMs: the C parameter

- The strength of regularization is determined by C
- Larger values of C: less regularization
  - Fit the training data as well as possible
  - Each individual data point is important to classify correctly
- Smaller values of C: more regularization – More tolerant of errors on individual data points

Reading List

- Concise explanation of linear regressions
  - https://www.analyticsvidhya.com/blog/2017/06/a-comprehensive-guide-for-linear-ridge-and-lasso-regression/
  - https://brunch.co.kr/@itschloe1/9

ANY QUESTIONS?